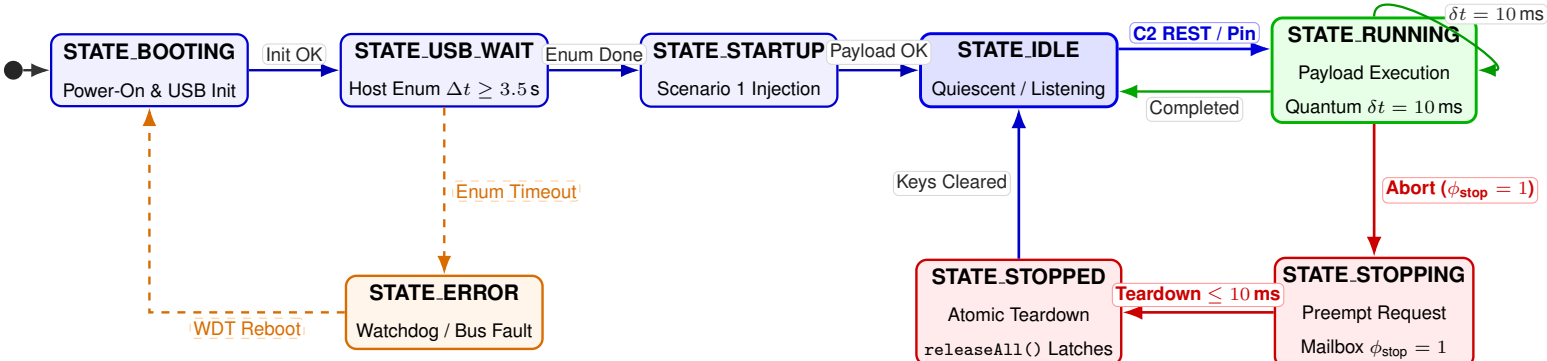
STATE_BOOTING
Power-On & USB Init
Init OK
STATE_USB_WAIT
Host Enum $\Delta t \geq 3.5\,\mathrm{s}$
Enum Done
STATE_STARTUP
Scenario 1 Injection
Payload OK
STATE_IDLE
Quiescent / Listening
C2 REST / Pin
STATE_RUNNING
Payload Execution
Quantum $\delta t = 10\,\mathrm{ms}$
$\delta t = 10\,\mathrm{ms}$
Completed
Enum Timeout
Keys Cleared
Abort ($\phi_{\mathrm{stop}} = 1$)
STATE_ERROR
Watchdog / Bus Fault
WDT Reboot
STATE_STOPPED
Atomic Teardown
releaseAll() Latches
Teardown $\leq 10\,\mathrm{ms}$
STATE_STOPPING
Preempt Request
Mailbox $\phi_{\mathrm{stop}} = 1$